

ASS 16.3

2303A52346

Python to C++ Conversion

Task:

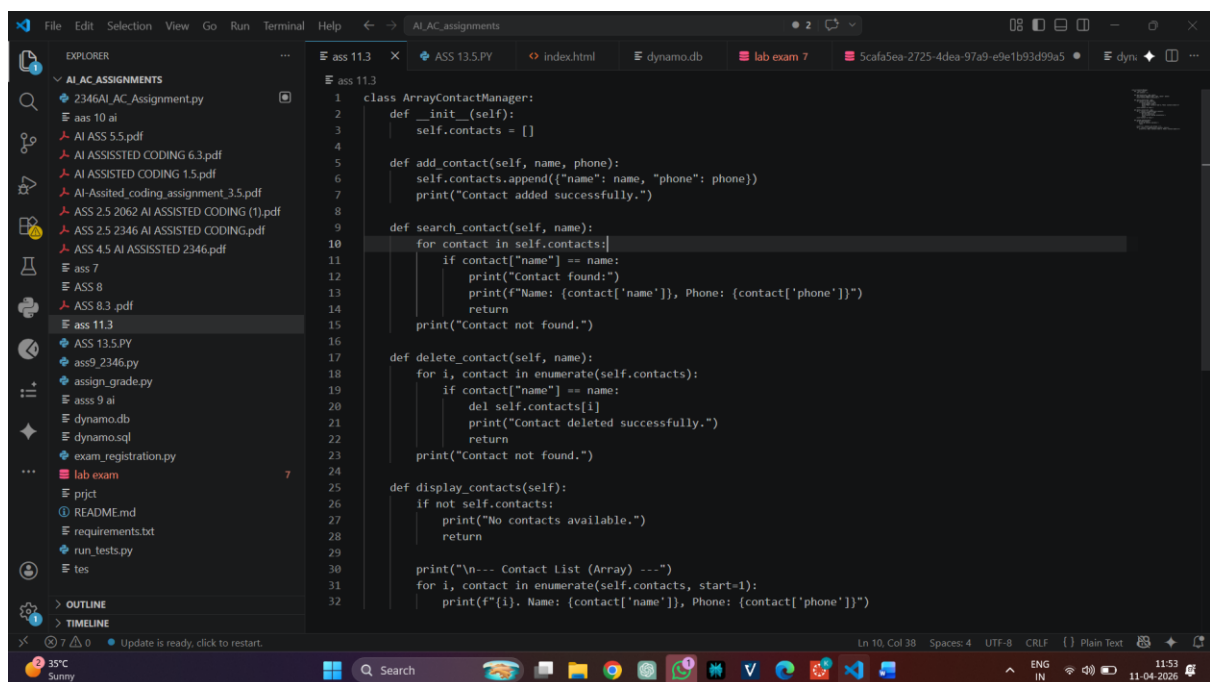
Translate a simple Python class into C++ using AI assistance.

Instructions:

- Prompt AI to convert a Python class representing a Student with attributes and methods into equivalent C++ code
- Ensure proper handling of:
 - o Constructors
 - o Data types
 - o Access specifiers
- Compile and run both versions to verify output consistency

Expected Output:

An equivalent working C++ class translated from Python.



```
1 class ArrayContactManager:
2     def __init__(self):
3         self.contacts = []
4
5     def add_contact(self, name, phone):
6         self.contacts.append({"name": name, "phone": phone})
7         print("Contact added successfully.")
8
9     def search_contact(self, name):
10        for contact in self.contacts:
11            if contact["name"] == name:
12                print("Contact found:")
13                print(f"Name: {contact['name']}, Phone: {contact['phone']}")
14                return
15        print("Contact not found.")
16
17    def delete_contact(self, name):
18        for i, contact in enumerate(self.contacts):
19            if contact["name"] == name:
20                del self.contacts[i]
21                print("Contact deleted successfully.")
22                return
23        print("Contact not found.")
24
25    def display_contacts(self):
26        if not self.contacts:
27            print("No contacts available.")
28            return
29
30        print("\n-- Contact List (Array) --")
31        for i, contact in enumerate(self.contacts, start=1):
32            print(f"{i}. Name: {contact['name']}, Phone: {contact['phone']}")
```

Java to Python Function Conversion

Task:

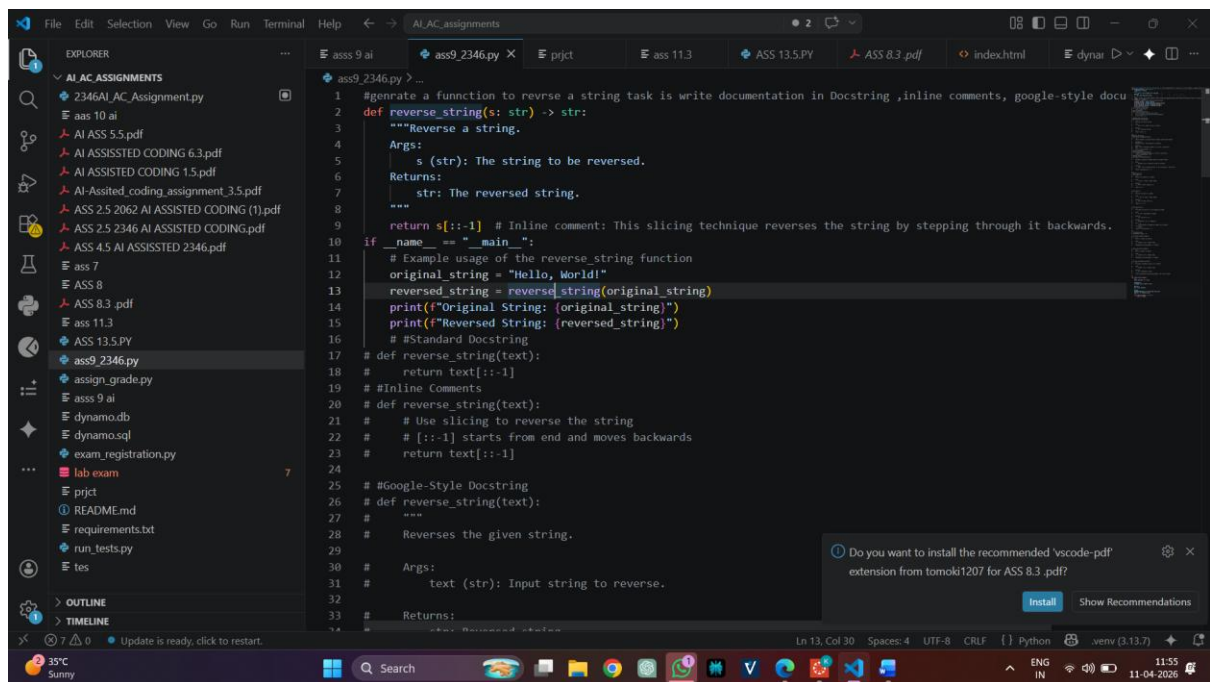
Convert a Java method that checks whether a number is prime into Python.

Instructions:

- Ask AI to translate a Java isPrime() method into Python
- Test the function with multiple inputs
- Ensure the Python version follows Pythonic syntax and logic

Expected Output:

A correct Python function equivalent to the original Java prime-checking method.



```
1 #generate a function to reverse a string task is write documentation in Docstring ,inline comments, google-style docu
2 def reverse_string(s: str) -> str:
3     """Reverse a string.
4     Args:
5         s (str): The string to be reversed.
6     Returns:
7         str: The reversed string.
8     """
9     return s[::-1] # Inline comment: This slicing technique reverses the string by stepping through it backwards.
10 if __name__ == "__main__":
11     # Example usage of the reverse string function
12     original_string = "Hello, World!"
13     reversed_string = reverse_string(original_string)
14     print(f"Original String: {original_string}")
15     print(f"Reversed String: {reversed_string}")
16     # Standard Docstring
17 # def reverse_string(text):
18 #     return text[::-1]
19 # #Inline Comments
20 # def reverse_string(text):
21 #     # Use slicing to reverse the string
22 #     # [::-1] starts from end and moves backwards
23 #     return text[::-1]
24
25 # Google-Style Docstring
26 # def reverse_string(text):
27 #     """
28 #     Reverses the given string.
29 #
30 #     Args:
31 #         text (str): Input string to reverse.
32 #
33 #     Returns:
34 #         str: Reversed string.
35 #     """
36 #     return text[::-1]
```

Do you want to install the recommended 'vscode-pdf' extension from tomoki1207 for ASS 8.3 .pdf?

Install Show Recommendations